



9 Fluorine-18 ($^{18}_{9}\text{F}$) decays by beta-plus (β^+) emission with a half-life of 110 minutes.

(a) (i) State the name of the beta-plus particle.

..... [1]

(ii) Show that the decay constant of fluorine-18 is $1.05 \times 10^{-4} \text{ s}^{-1}$.

[1]

(iii) Determine the activity of $2.1 \times 10^{-12} \text{ kg}$ of fluorine-18.

activity = Bq [3]

(b) A small sample of fluorine-18 injected into the body acts as a tracer for use in medical imaging.

(i) Describe how the interaction of a β^+ particle with an electron in the body enables the formation of an image.

.....

 [3]

(ii) Suggest why 110 minutes is a suitable half-life for a nuclide used as a tracer in medical diagnosis.

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 [2]

[Total: 10]

